

FINAL PROGRESS REPORT

TECHNICAL ASSESSMENT & COMPLETED DELIVERABLES

Project:	AI Research Paper Assistant	Date:	2026-05-30
Target:	Banao Technologies Assessment	Status:	Production-Ready (100% Completed)

1. Project Overview

A RAG-powered PDF assistant built with a Flask backend and a modern violet dark-theme UI. It enables users to upload multiple documents, query sections using smart lexical chunking (via Groq Llama-3.3), get automated structured summaries, export APA/BibTeX citations, and study key concepts using 3D flippable flashcards.

2. Delivered Deliverables & Achievements

- **Backend Framework Setup:** Built the complete Flask application (app.py) with session management and disk-cached text files to avoid session capacity limits.
- **PDF Text Extraction:** Integrated PyMuPDF (fitz) to scrape text and count words in real-time, with scanned-PDF image audit warnings.
- **Smart Lexical Chunking & RAG:** Created an overlapping sliding-window chunker (3,000 words, 500-word overlap) for files over 6,000 words, paired with a lexical keyword overlap matcher to send optimal context to Groq.
- **Structured Auto-Summary:** Developed structured AI templates dividing academic papers into One-Line Summaries, Contributions, Methodologies, and Main Results.
- **Frontend UI/UX Interface:** Styled a premium violet dark-mode dashboard (index.html) with custom drag-and-drop zones, tab selectors, and responsive chat timeline layouts.
- **Interactive 3D Study Flashcards:** Programmed an AI concept extractor and styled hardware-accelerated CSS 3D perspectives to render digital study cards that flip 180 degrees on click.
- **Academic Citation Generator:** Programmed automated APA style and LaTeX BibTeX citation formatting with copy-to-clipboard JS triggers.
- **Fail-Safe Parsing & Layouts:** Constrained layouts to 100vh height to fix timeline scrolling. Applied regex filters (re.search) to cleanly parse LLM JSON responses.

3. Current Status

The technical development phase is **100% completed**. The system has been fully built, integrated, syntactically verified, and successfully stress-tested. It is fully operational, stable, and ready for production launch.

4. Next Steps (Post-Completion Handover Roadmap)

- **Key Activation:** Provide the Banao team with environment variables for the Groq API key configuration.
- **Production Deployment:** Host the Flask application on a scalable cloud platform (e.g., Render, Heroku, or AWS Elastic Beanstalk).
- **Code Handover:** Transfer the git code repository to the Banao Technologies engineering team for evaluation.